



deMITasse Energies team with Vijay Prateik, Founder and CEO (2nd last from left), and Megha Rawat, Co-founder (right most)

a zero-emission technology. It can both generate and store electricity with high efficiency.

deMITasse's power plant works on a fundamentally similar principle as a thermal power plant that generates electricity by moving a turbine using steam. But unlike a regular thermal plant, the company's plant uses a different type of working fluid to power a different type of thermodynamic cycle, which offers better efficiency and has better pressure and temperature characteristics than water.

Vijay Prateik, the Founder and CEO, says, "Our plants utilise available external energy sources like waste heat from industries, solar thermal, off-peak electricity, etc to capture energy within our proprietary working fluid. This captured energy can be converted into usable power by passing this energy-rich (high-pressure, high-temperature) working fluid through a turbogenerator."

The secret sauce of the company's power plant is the process of re-liquefying the working fluid after it has been used for moving the turbogenerator. deMITasse Energies' power plant utilises complex chemical and catalytic processes to liquefy the working fluid, thus getting rid of

the energy-inefficient components like condensers and compressors used in the existing thermal plants and making it an energy-efficient process.

Thanks to their "highly energetic" chemicals and catalysts their power plant can also double as an energy storage system, when the energy demand is low, or can be used in conjunction with solar farms for storing energy. Prateik says, "The round-trip efficiency of our plant is about 70-85% as compared to about 30-36% of thermal power plants, 38-40% of advanced super-critical power plants, and about 5-10% of ORC plants."

The working fluid and the chemical processes enable the plant to convert and use low-grade waste thermal energy that is widely available and unutilised by the existing thermal power plants. "Current technologies are prohibitively expensive and can extract power only from sources that are above 150-200°C, while our technology can economically utilise thermal sources at temperatures as low as 70-80°C, and it is in this 80-120°C region where the majority of waste heat resides," Prateik informs.

This technology can also be used as grid-scale energy storage plants.

Unlike existing energy storage systems that use electrochemical batteries, such as the lithium-ion cells, the company's technology works on its thermochemical cycle that exploits the thermal energy exchange that occurs during high-energy chemical reactions. Prateik says, "Unlike electrochemical batteries, no potential needs to be maintained to generate power, thereby eliminating the issue of energy dissipation and giving our plants the ability to store energy for several months without any energy loss."

The deMITasse Energies' power plant offers several advantages over other existing energy generation technologies. The low-cost technology is extremely efficient and can store energy for up to 48 months with almost no loss. Moreover, the technology works on a closed loop cycle with no combustion of any fuel—resulting in zero emissions operations. The technology is highly scalable and can be deployed for a variety of applications ranging from 10kW to 2000MW, with an energy storage capacity of 50kWh to 20GWh.

"Our systems are modular and highly automated, thus providing high upgradability as the needs of the customer change. There are several applications and use cases of our technology, from grid-scale energy storage plants to industrial waste heat recovery systems, from military applications to spacecraft propulsion outside the earth's atmosphere," declares the CEO.

The company is in the process of setting up a pilot plant in Denver, Colorado. The company is also planning to expand its core team and raise more funds in the coming times. However, deMITasse Energies' long-term goal is to build grid-interacting power plants and sell energy on a subscription basis to its customers. **EFY**

—Sharad Bhowmick